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(1) Research Program/Project/Study Title: Effects of Drying Temperature on the Pollen Viability of Coconut (<i>Cocos nucifera</i> L.) Using Various Growth Media	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Coconut	(8) Discipline/Field of Specialization Agriculture

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Crop and Soils Laboratory of Capiz State University, Burias Campus, Mambusao, Capiz. It was laid out using the Completely Randomized Design (CRD) with two factors, wherein Factor A consisted of different drying temperatures: A1 (40°C) and A2 (45°C). Factor B involved different culture media: B1 (potato dextrose agar) and B2 (platelet count agar). Each treatment was replicated three times. Differences among treatment means were compared using the Least Significant Difference (LSD) test, interpreted at a 5% level of significance. The study aimed to determine the effects of drying temperature on the pollen viability of coconut using various growth media.

Analysis of Variance (ANOVA) indicated that the drying temperature did not significantly affected the pollen viability. Growth media significantly affected pollen viability. Using platelet count agar resulted in a higher viability rate than potato dextrose agar. Grain with a spheroidal shape and a white to light brown tint indicated germinated pollen, while grain with a spheroidal shape and a dark brown to black tint indicated ungerminated pollen. The length of the spheroidal grain ranged from 26 to 43 µm when dried at 40°C and from 32 to 45 µm when dried at 45°C, as shown in Figures 2, 3, 4, and 5.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Pollen viability test, Pollen viability, coconut, coconut pollen, culture media

- 1. Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Coconut (*Cocos nucifera* L.) is a vital crop in tropical regions, providing a range of products from food to industrial materials. The reproductive success of coconut plants, and consequently their yield, is heavily dependent on the viability of their pollen. Pollen viability is crucial for successful fertilization and fruit set, making it an important focus for researchers and cultivators. One of the significant factors affecting pollen viability is the drying temperature, which can vary widely in different environments and handling procedures.

Pollen viability refers to the ability of pollen grains to germinate and effect fertilization, a critical aspect of plant reproductive biology (Shivanna & Johri, 1985). It is influenced by both intrinsic factors, such as genetic traits and the physiological state of the pollen, and extrinsic factors, including environmental conditions during collection, drying, and storage (Stanley & Linskens, 1974). Temperature, particularly during the drying process, is a key extrinsic factor that can significantly affect pollen viability.

Drying temperature plays a crucial role in maintaining or reducing the viability of pollen. High temperatures can cause dehydration and damage to pollen grains, leading to a loss of viability (Heslop-Harrison & Heslop-Harrison, 1985). Conversely, low temperatures can slow down metabolic processes and preserve pollen viability but may not be practical in tropical climates where coconuts are commonly grown. Therefore, identifying an optimal drying temperature that balances these effects is vital for maintaining high pollen viability.

Research has shown that the optimal drying temperature for pollen viability varies among different plant species and even among varieties within a species. For instance, studies on pollen viability in cereals and other crops have highlighted that moderate drying temperatures (20-30°C) tend to preserve viability better than higher or lower extremes (Hoekstra, 1995). Specific research on coconut pollen viability has also indicated that temperatures above 40°C can significantly reduce germination rates and pollen viability (Perera et al., 2009).

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The interaction between drying temperature and other factors, such as humidity and storage duration, can compound the effects on pollen viability. Research by Baker and Baker (1979) demonstrated that the combination of high temperature and low humidity during drying could lead to rapid dehydration and reduced pollen viability. Conversely, maintaining a controlled environment with moderate temperature and humidity during drying and storage has been shown to preserve pollen viability more effectively (Shivanna & Johri, 1985).

The implications of optimizing drying temperature for coconut pollen viability are significant for breeding programs and commercial coconut production. High pollen viability ensures successful fertilization, leading to higher fruit set and yield. This is particularly important for hybridization efforts aimed at developing new coconut varieties with improved traits, such as disease resistance, drought tolerance, and higher productivity (Samosir et al., 2016). Additionally, maintaining high pollen viability during storage and transportation is crucial for large-scale breeding programs and cross-regional germplasm exchange (Perera et al., 2009).

One important aspect affecting the viability of coconut pollen is the drying temperature, together with the kind and ratio of growing medium employed. Achieving the ideal drying temperature can boost fertilization rates, boost the effectiveness of coconut breeding operations, and eventually increase coconut harvests. Further study in this field will yield important information for creating efficient handling and storage procedures for pollen, enhancing the productivity and sustainability of coconut farming.

Objectives of the Study

Generally, the study aimed to investigate the effects of drying temperature on the pollen viability of coconut (*Cocos nucifera* L.) using various growth media. Specifically, the goals of the study were to:

1. Determine the viability of pollen as influenced by drying temperature.
2. Quantify the viability of coconut pollen using different growth media.
3. Examine the interaction effect between drying temperature and growth media.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Experimental Method

The study was conducted at the Crop and Soils Laboratory of Capiz State University, Burias Campus, Mambusao, Capiz. It was laid out using the Completely Randomized Design (CRD) with two factors, wherein Factor A consisted of different drying temperatures: A1 (40°C) and A2 (45°C). Factor B involved different culture media: B1 (potato dextrose agar) and B2 (platelet count agar). Each treatment was replicated three times. Differences among treatment means were compared using the Least Significant Difference (LSD) test, interpreted at the 5% level of significance. The study used the ImageJ application to determine the viability of pollen (Figure 1). The data gathered were analyzed using the Statistical Tool for Agricultural Research (STAR) and were interpreted at the 5% level of significance.

Cultural Practices and Management

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Preparation of coconut pollen

Coconut pollen was collected from the male flowers of Tagnanan Tall. Stripped male flowers were then weighed to measure the total pollen collected. Pollen was dried at 40°C (A1) for 30.5 hours and at 45°C (A2) for 21 hours. Cracking was performed after the pollen had been dried and withdrawn from the oven dryer using a wooden rolling pin to release the pollen grains. Lastly, the pollen sieve was tested for its viability.

Preparation of growth media

The culture media were prepared by heating 100 mL of distilled water. After that, 1 mL of boric acid solution, 10 g of sucrose, and 0.5 g of agar were added, stirring constantly until the ingredients were completely dissolved. Both potato dextrose agar and platelet count agar were used in this study.

Pollen Viability Test

Pollen viability tests were determined by preparing various growth media where pollen was cultivated. The agars used in the study were potato dextrose agar (B1) and platelet count agar (B2).

Pollen viability test were determined by:

$$\% \text{ Pollen Germination} = \frac{\text{Number of germinated pollen}}{\text{Total number of pollen counted}} \times 100$$

(11) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Pollen viability test result

Table 1 presents the effects of drying temperature on the pollen viability of coconut (*Cocos nucifera* L.) using various growth media. Analysis of Variance (ANOVA) indicated that the result for drying temperature was not significant. This means that the drying temperature did not affect the viability of the pollen. Means ranged from 28.92% to 29.92%.

With regards to the effect of growing media used in pollen viability test, the ANOVA revealed a significant result. Platelet count agar (29.66%) demonstrated a higher viability rate than potato dextrose agar (28.18%).

Data revealed that there was no interaction effect between drying temperature and growth media.

Coconut pollen morphology observation

A trinocular compound microscope at 4X magnification was used to examine some of the morphological features of coconut pollen. Grain with a spheroidal shape and a white to light brown tint indicated germinated pollen. A pollen tube was present when there was active germinated pollen. Grain with a spheroidal shape and a dark brown to black tint indicated ungerminated pollen. The length of spheroidal grains ranged from 26 to 43 µm when dried at 40°C and from 32 to 45 µm when dried at 45°C, as shown in Figures 2, 3, 4, and 5.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations

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are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions were reached:

1. Pollen viability of was not affected by drying temperature.
2. Growing media used in pollen viability test significantly influenced pollen germination.
3. There was no interaction effect between drying temperature and culture media.

Recommendations

Based on the results of this study, the following recommendations are forwarded:

1. Use platelet count agar doing pollen viability test for coconut pollen.
2. Conduct similar studies with the following modifications:
 - a. explore other drying temperature for higher pollen viability
 - b. use other coconut varieties in pollen testing
 - c. enhance growth media
 - d. utilize other agar in media preparations

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

References

- Baker, H. G., & Baker, I. (1979). Starch in angiosperm pollen grains and its evolutionary significance. *American Journal of Botany*, 66(5), 591-600.
- Heslop-Harrison, J., & Heslop-Harrison, Y. (1985). Germination of pollens on nutrient media: Pollen-stigma interaction and subsequent fertilization. *Journal of Cell Science*, 2, 1-15.
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- Jayasekara, C., Fernando, S. C., & Ranasinghe, C. S. (2002). Factors affecting in vitro pollen germination in coconut (*Cocos nucifera* L.). *Ceylon Journal of Science (Biological Sciences)*, 30, 11-19.
- Perera, L., Russell, J. R., & Powell, W. (2009). Identification of quantitative trait loci for yield and yield components in coconut (*Cocos nucifera* L.). *Euphytica*, 168, 69-80.
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APPENDICES

Table 1. Pollen viability (%) of coconut (*Cocos nucifera L.*) using various growth media as affected by drying temperature.

Viability of Pollen (%)				
Treatment	Growth Media *			
Degree of Drying ^{ns}	B1 – Potato Dextrose Agar	B2 – Platelet Count Agar	Total	Mean
A1 – 40°C at	28.18	29.66	57.84	28.92
A2 – 45°C	28.48	29.97	58.45	29.22
Total	56.66	59.63		
Mean	28.33 ^b	29.82 ^a		

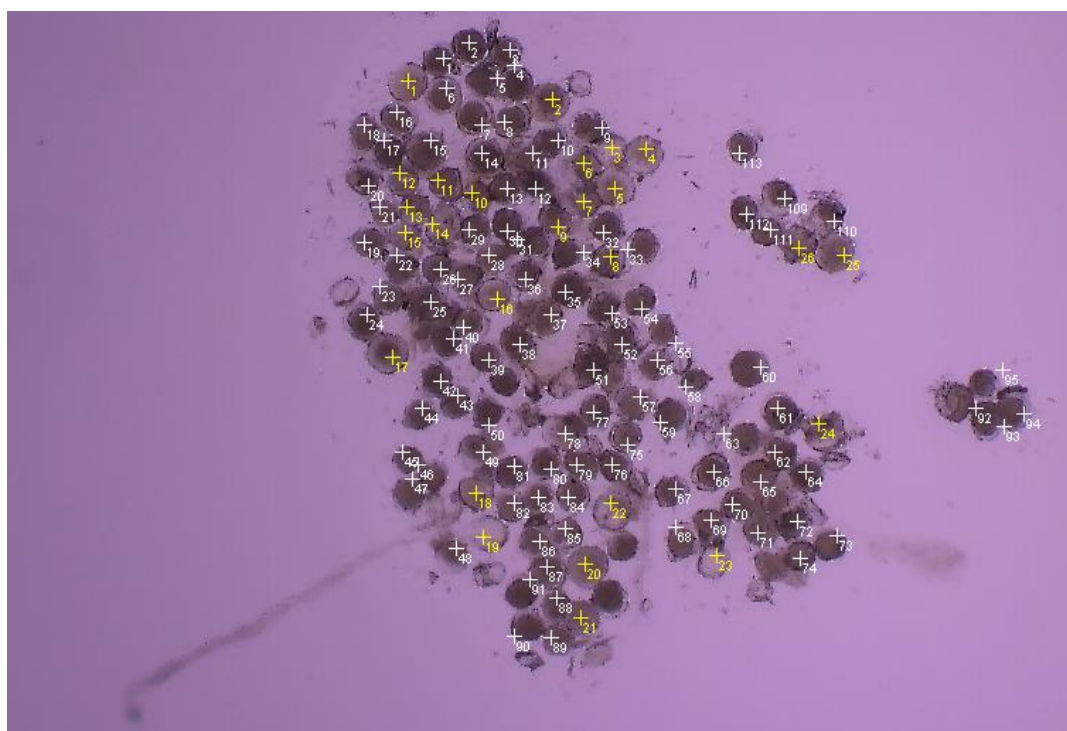


Figure 1. Counting of pollen using image J application: Yellow cross pointed tool indicates germinated pollen; white cross pointed tool indicates ungerminated pollen.

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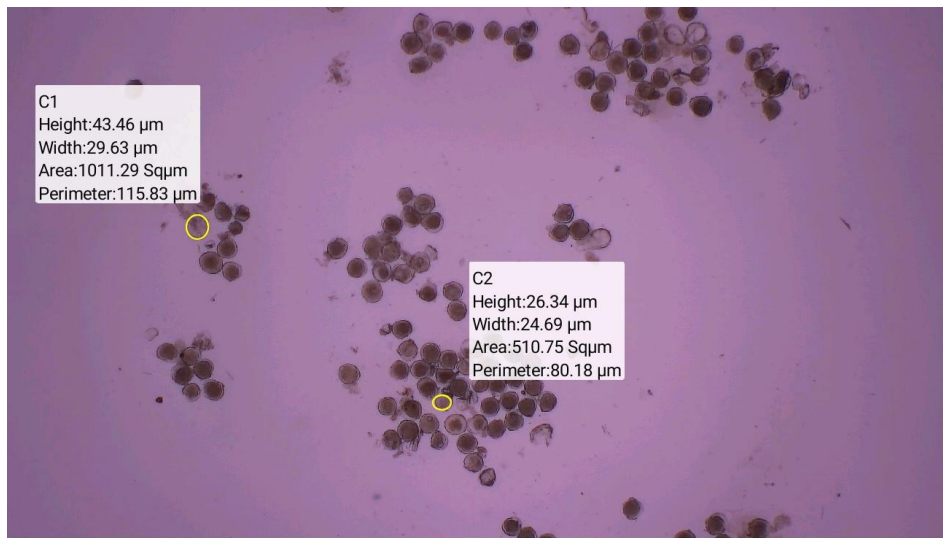


Figure 2. Measurements of Tagnanan Tall pollen (largest – C1 and smallest – C2) dried at 40°C using platelet count agar.

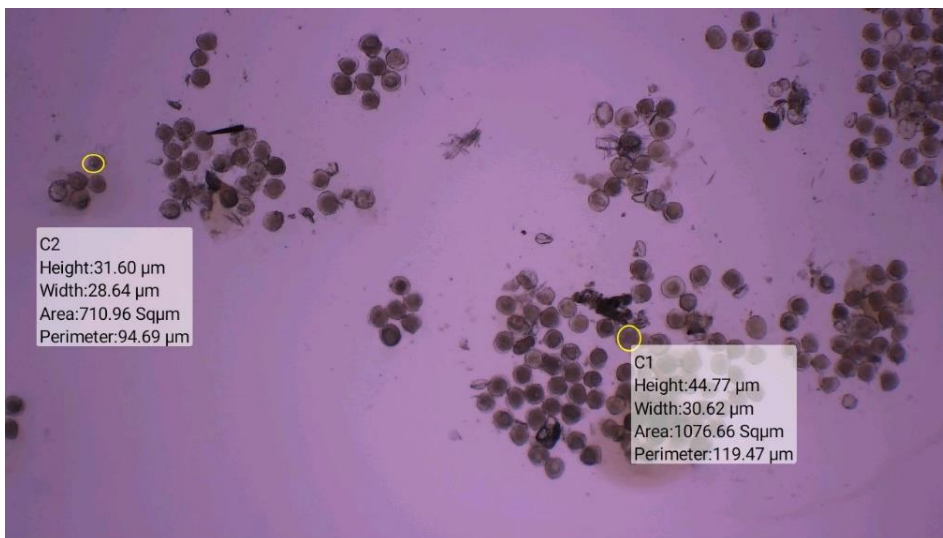


Figure 3. Measurements of Tagnanan Tall pollen (largest – C1 and smallest – C2) dried at 45°C using platelet count agar.

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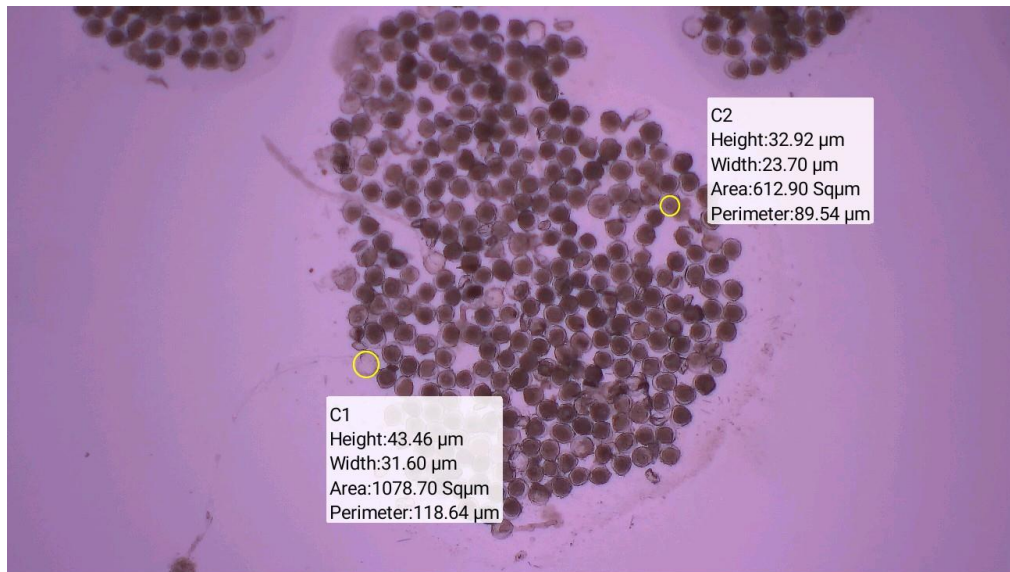


Figure 4. Measurements of Tagnanan Tall pollen (largest – C1 and smallest – C2) dried at 40°C using potato dextrose agar.

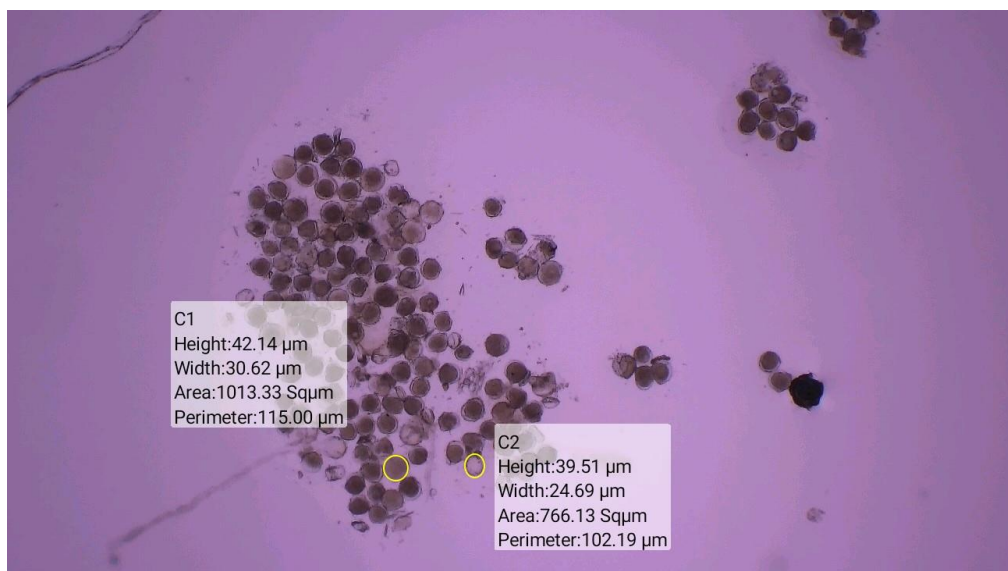


Figure 5. Measurements of Tagnanan Tall pollen (largest – C1 and smallest – C2) dried at 45°C using potato dextrose agar.